

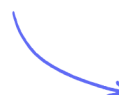
Calculator Questions

Circle Theorems

Angles at Centre & Circumference / Angle in a Semicircle / Theorems with Chords & Tangents / Angles in Cyclic Quadrilaterals / Angles in the Same Segment / The Alternate Segment Theorem

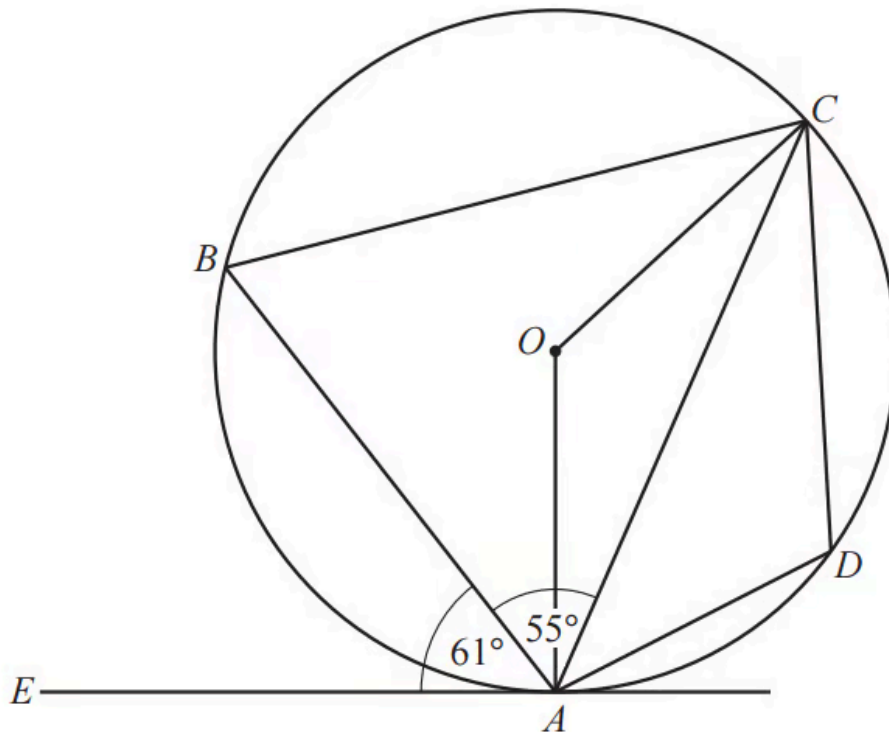
Medium (6 questions)	/26
Hard (8 questions)	/41
Very Hard (5 questions)	/21
Total Marks	/88

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Medium Questions

1 (a)



NOT TO
SCALE

In the diagram, A , B , C and D lie on the circle, centre O .

EA is a tangent to the circle at A .

Angle $EAB = 61^\circ$ and angle $BAC = 55^\circ$.

Find angle BAO .

Angle $BAO = \dots\dots\dots$

(1 mark)

(b) Find angle AOC .

Angle $AOC = \dots\dots\dots$

(2 marks)

(c) Find angle ABC .

Angle $ABC = \dots\dots\dots$

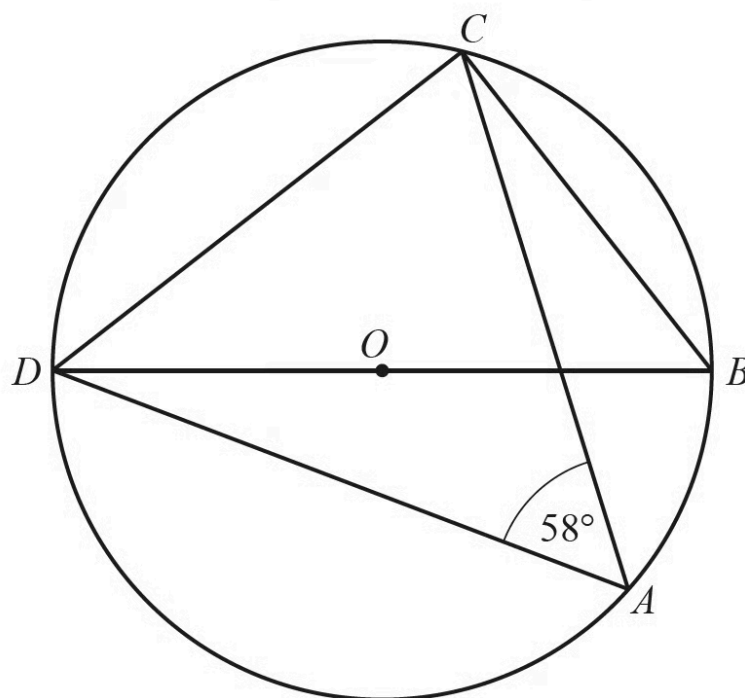
(1 mark)

(d) Find angle CDA .

Angle $CDA = \dots\dots\dots$

(1 mark)

2



NOT TO
SCALE

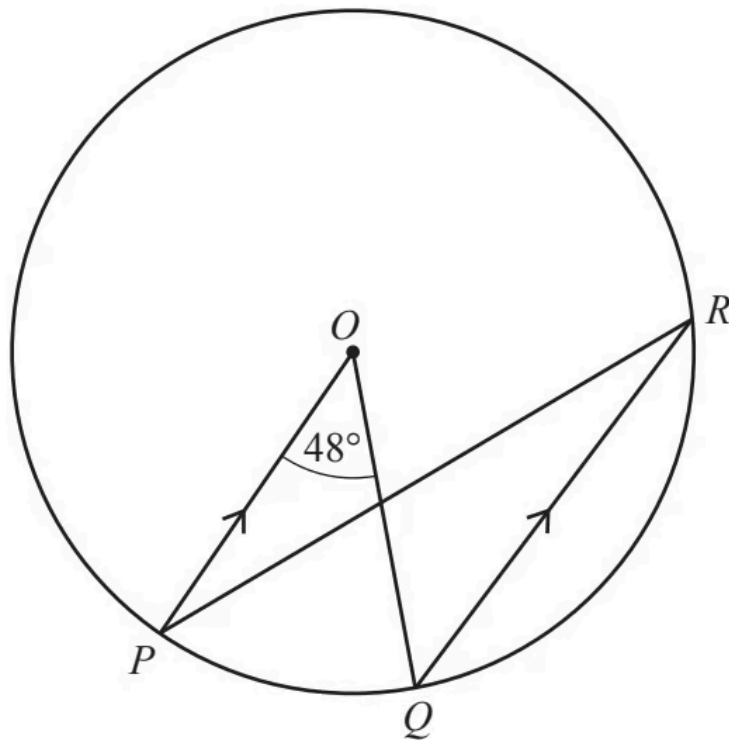
A , B , C and D are points on the circumference of the circle, centre O . DOB is a straight line and angle $DAC = 58^\circ$.

Find angle CDB .

Angle $CDB = \dots\dots\dots$

(3 marks)

3



NOT TO
SCALE

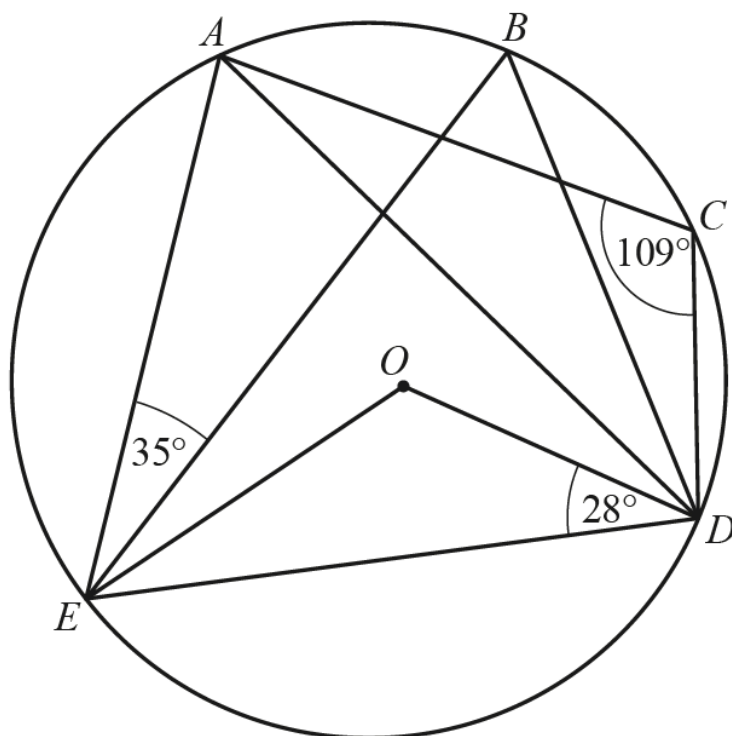
P , Q and R are points on the circumference of the circle, centre O .
 PO is parallel to QR and angle $POQ = 48^\circ$.

Find angle OPR .

Angle $OPR = \dots\dots\dots$

(2 marks)

4 (a)



NOT TO
SCALE

A, B, C, D and E lie on the circle, centre O .

Angle $AEB = 35^\circ$, angle $ODE = 28^\circ$ and angle $ACD = 109^\circ$.

Work out the following angles, giving reasons for your answers.

i) Angle $EBD = \dots\dots\dots$ because $\dots\dots\dots$

[3]

ii) Angle $EAD = \dots\dots\dots$ because $\dots\dots\dots$

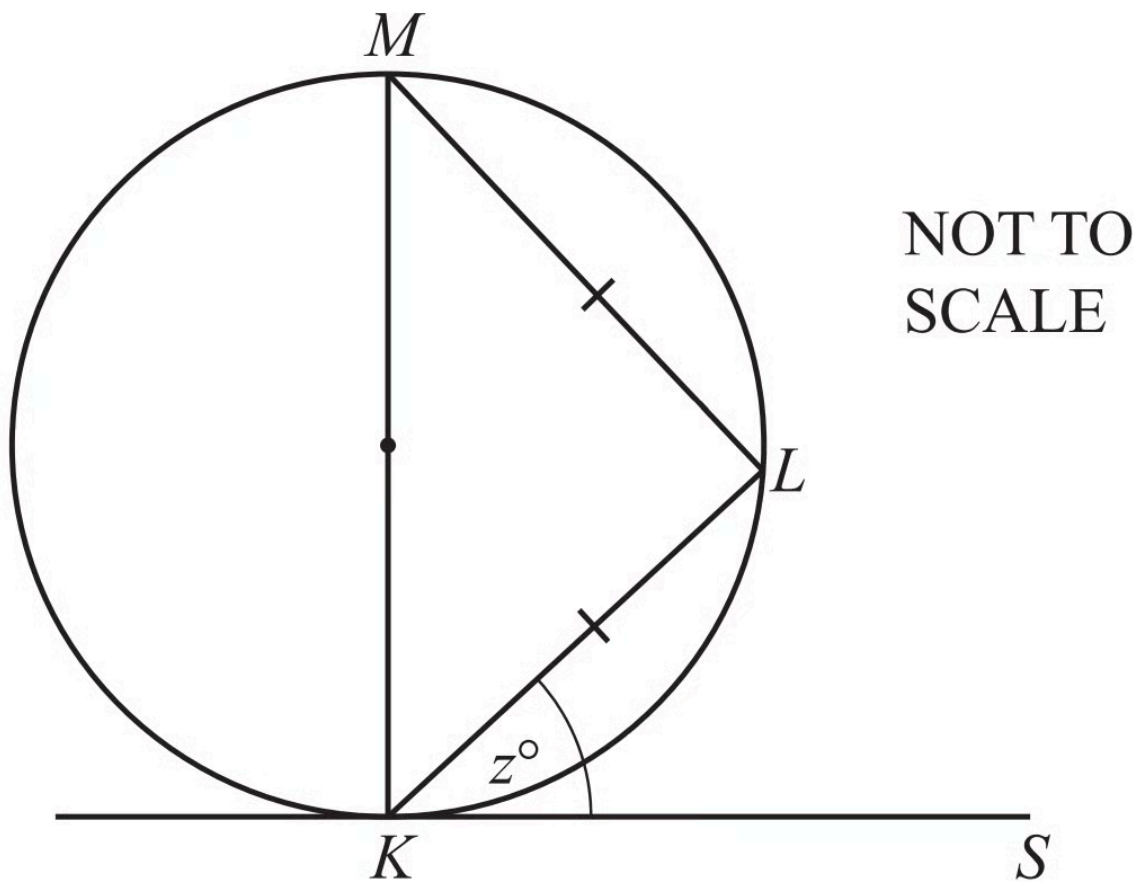
[2]

(5 marks)

(b) Work out angle BEO .

Angle $BEO = \dots\dots\dots$

5



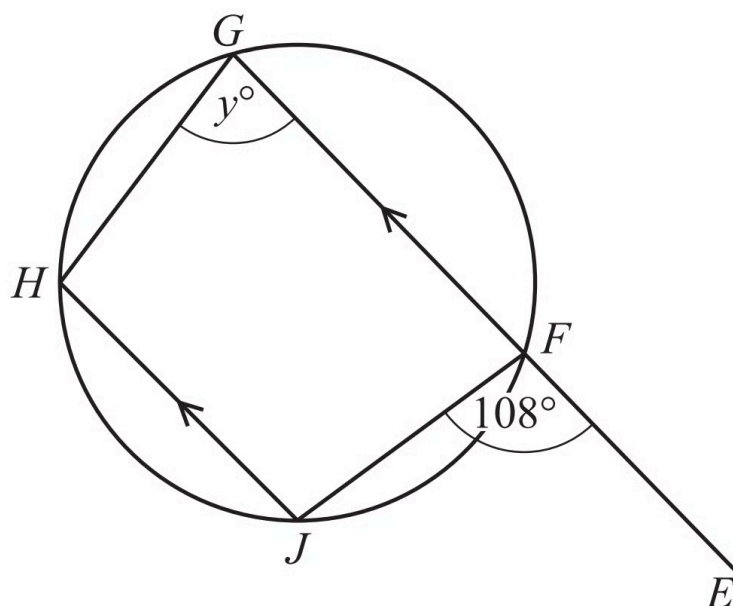
K , L and M are points on the circle.
 KS is a tangent to the circle at K .
 KM is a diameter and triangle KLM is isosceles.

Find the value of z .

$z = \dots\dots\dots$

(2 marks)

6

NOT TO
SCALE

F , G , H and J are points on the circle.
 EFG is a straight line parallel to JH .

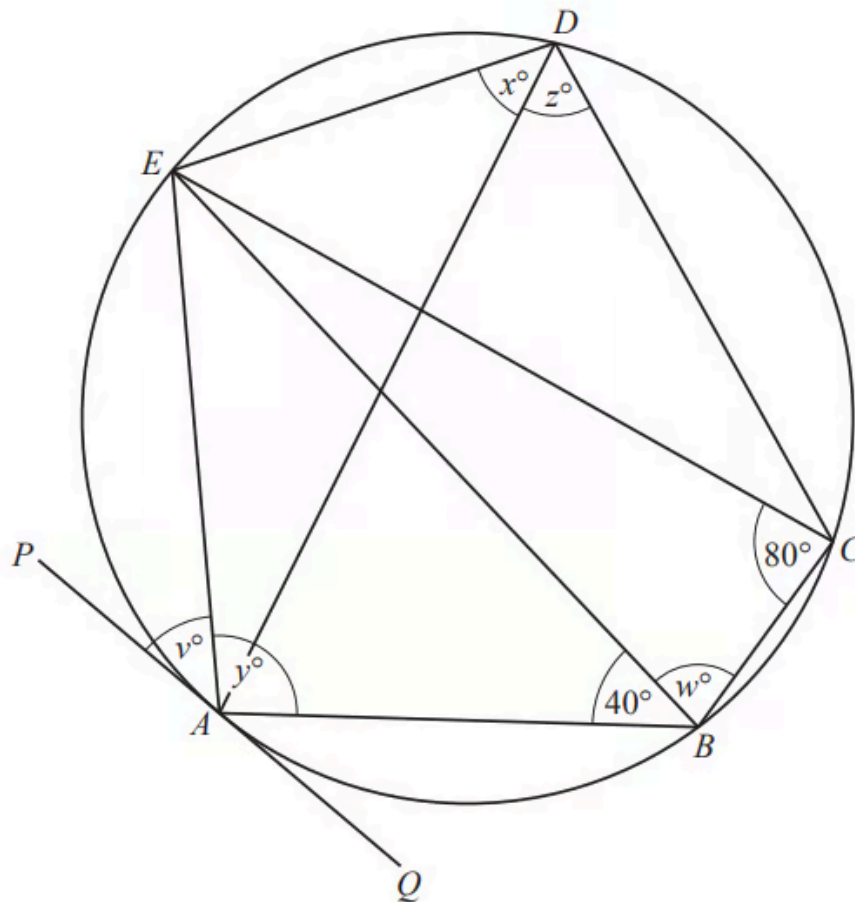
Find the value of y .

$y =$

(6 marks)

Hard Questions

1

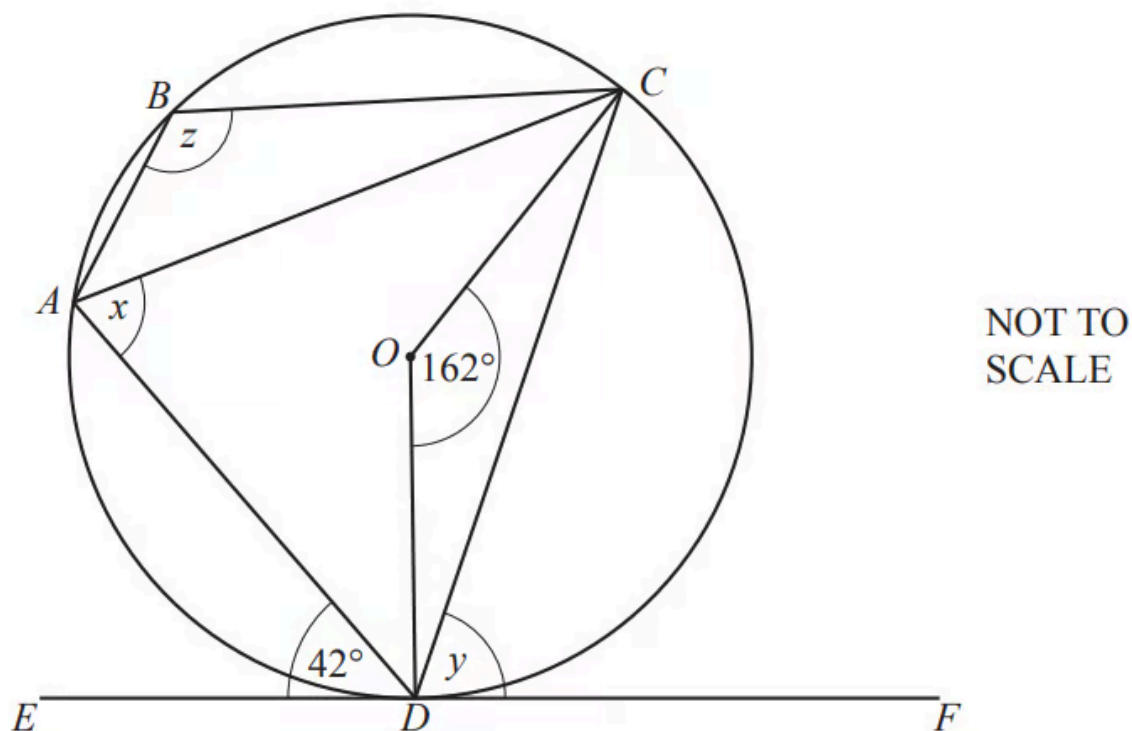


NOT TO
SCALE

The points A, B, C, D and E lie on the circle.
 PAQ is a tangent to the circle at A and $EC = EB$.
 Angle $ECB = 80^\circ$ and angle $ABE = 40^\circ$.

Find the values of v , w , x , y and z .

$v = \dots\dots\dots$ $w = \dots\dots\dots$ $x = \dots\dots\dots$ $y = \dots\dots\dots$ $z = \dots\dots\dots$
 (5 marks)



A, B, C and D are points on the circle, centre O .

EF is a tangent to the circle at D .

Angle $ADE = 42^\circ$ and angle $COD = 162^\circ$.

Find the following angles, giving reasons for each of your answers.

i) Angle x

$x = \dots\dots\dots$ because $\dots\dots\dots$ [2]

ii) Angle y

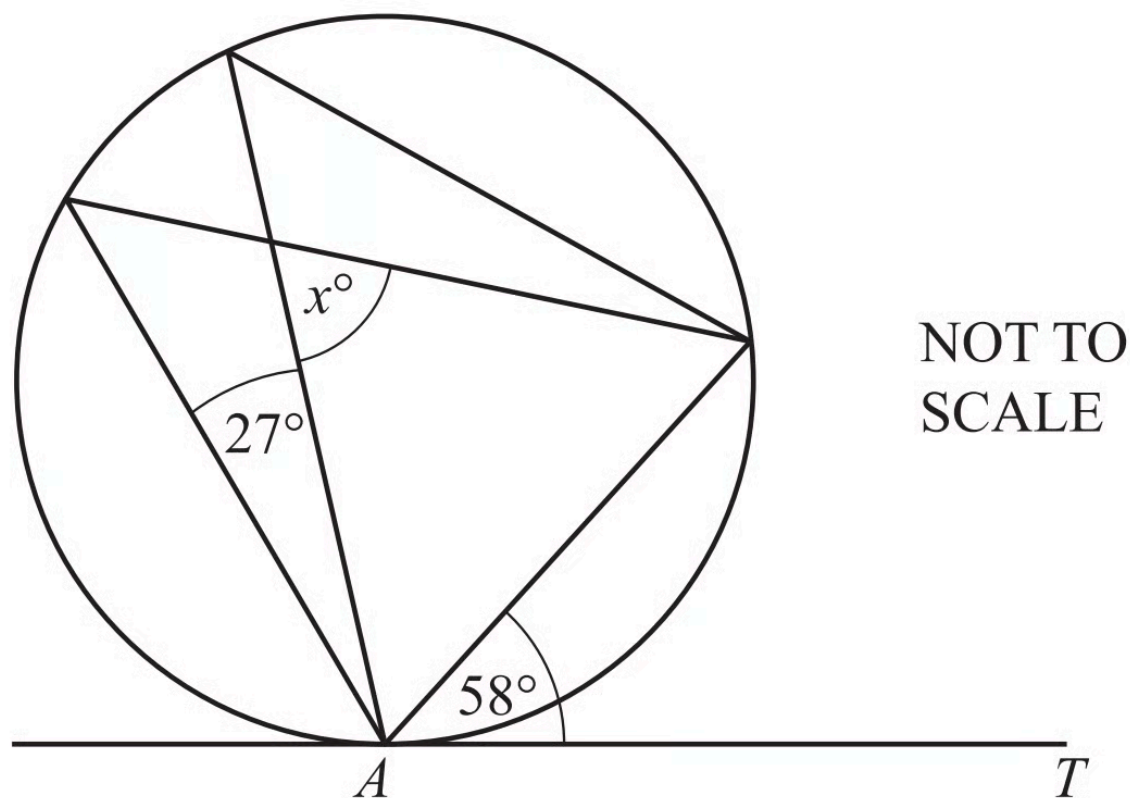
$y = \dots\dots\dots$ because $\dots\dots\dots$ [2]

iii) Angle z

$z = \dots\dots\dots$ because $\dots\dots\dots$ [3]

(7 marks)

3

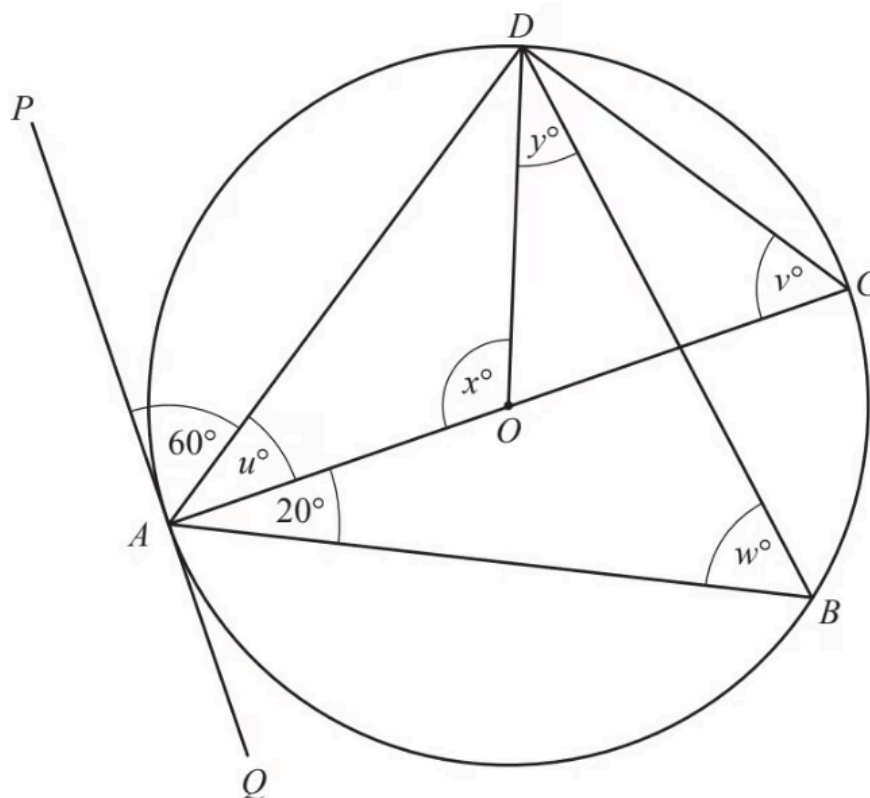


AT is a tangent to the circle at A .

Find the value of x .

$x = \dots\dots\dots$

(2 marks)



NOT TO
SCALE

A, B, C and D lie on the circle, centre O , with diameter AC .

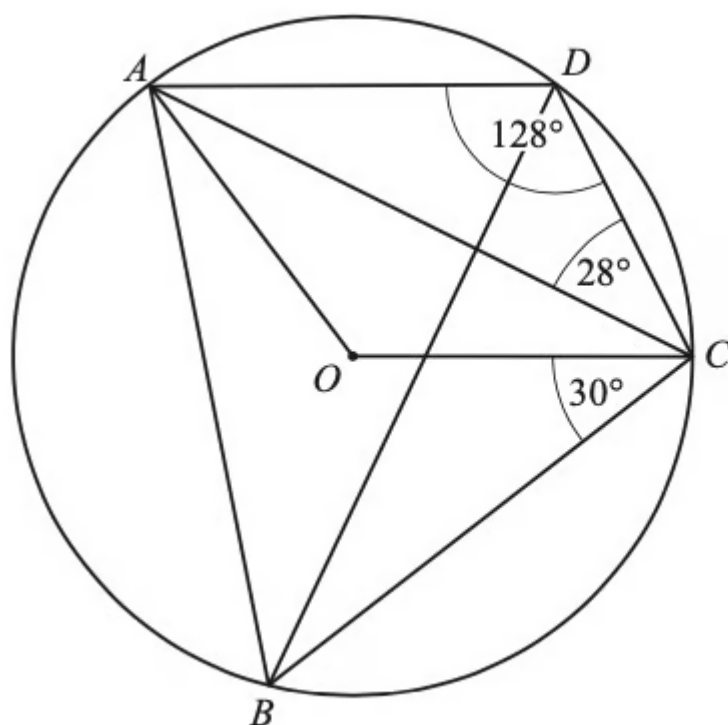
PQ is a tangent to the circle at A .

Angle $PAD = 60^\circ$ and angle $BAC = 20^\circ$.

Find the values of u , v , w , x and y .

$u = \dots\dots\dots$, $v = \dots\dots\dots$, $w = \dots\dots\dots$, $x = \dots\dots\dots$, $y = \dots\dots\dots$
(6 marks)

5 (a)



NOT TO
SCALE

In the diagram, A, B, C and D lie on the circle, centre O .
Angle $ADC = 128^\circ$, angle $ACD = 28^\circ$ and angle $BCO = 30^\circ$.

Show that obtuse angle $AOC = 104^\circ$.

Give a reason for each step of your working.

(3 marks)

(b) i) Find angle BAO .

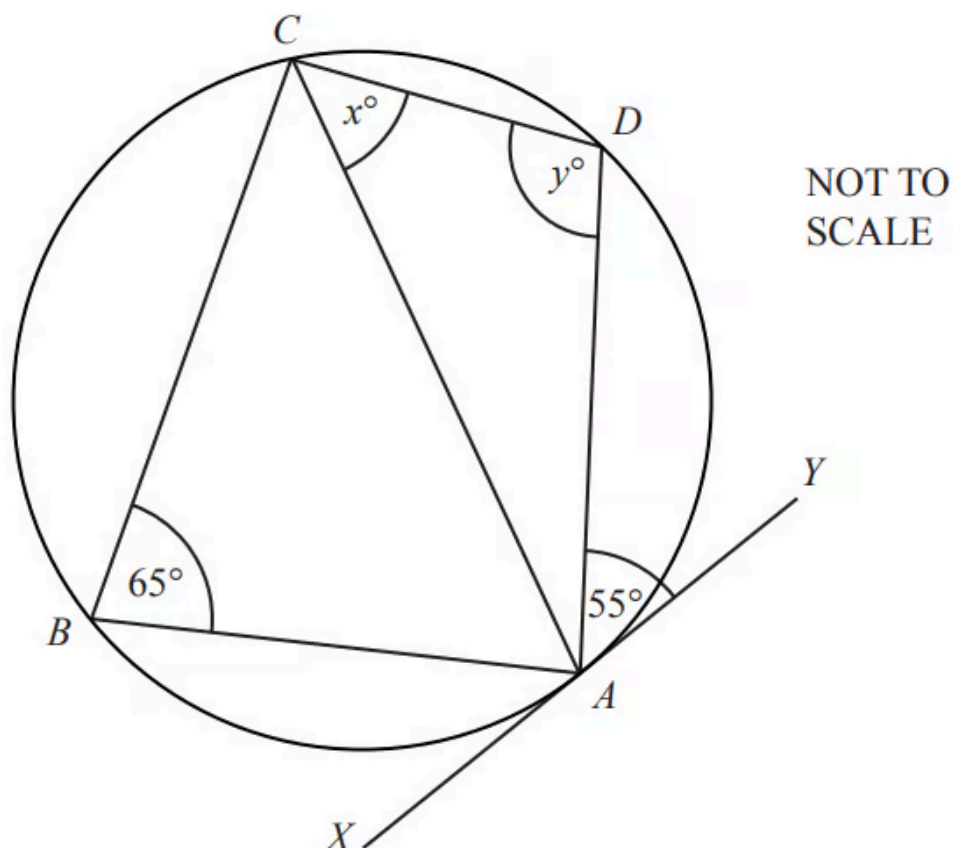
Angle $BAO = \dots\dots\dots$ [2]

ii) Find angle ABD .

Angle $ABD = \dots\dots\dots$ [1]

(3 marks)

6 (a)



A , B , C and D are points on the circumference of the circle.
The line XY is a tangent to the circle at A .

Find the value of x , giving a reason for your answer.

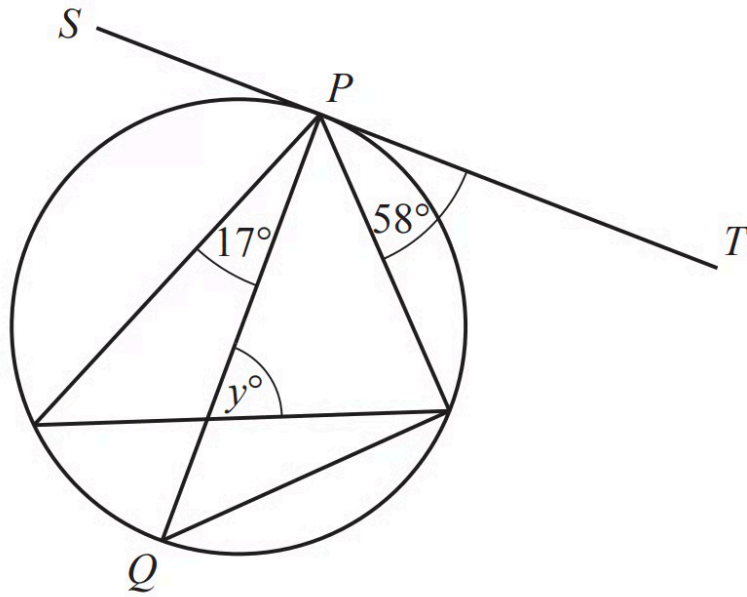
$x = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

(b) Find the value of y , giving a reason for your answer.

$y = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)



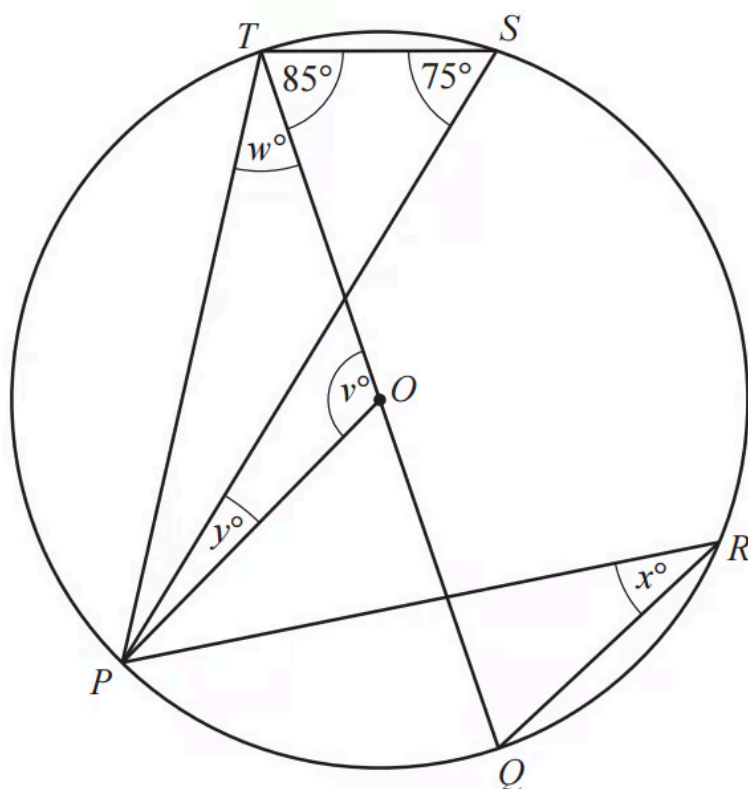
NOT TO
SCALE

The diagram shows a circle with diameter PQ .
 SPT is a tangent to the circle at P .

Find the value of y .

$y = \dots\dots\dots$

(5 marks)



NOT TO
SCALE

P, Q, R, S and T lie on the circle, centre O .
Angle $PST = 75^\circ$ and angle $QTS = 85^\circ$.

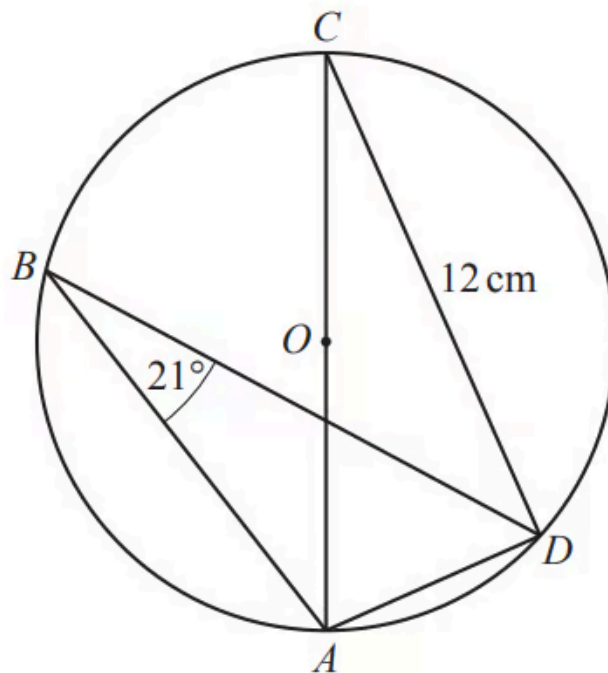
Find the values of v , w , x and y .

$v =$
 $w =$
 $x =$
 $y =$

(6 marks)

Very Hard Questions

1



NOT TO
SCALE

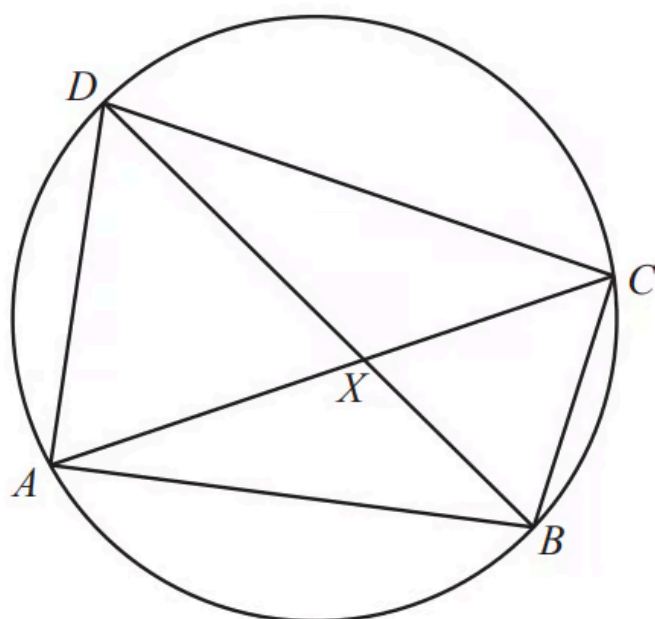
A , B , C and D are points on the circle, centre O .
Angle $ABD = 21^\circ$ and $CD = 12$ cm.

Calculate the area of the circle.

..... cm^2

(5 marks)

2



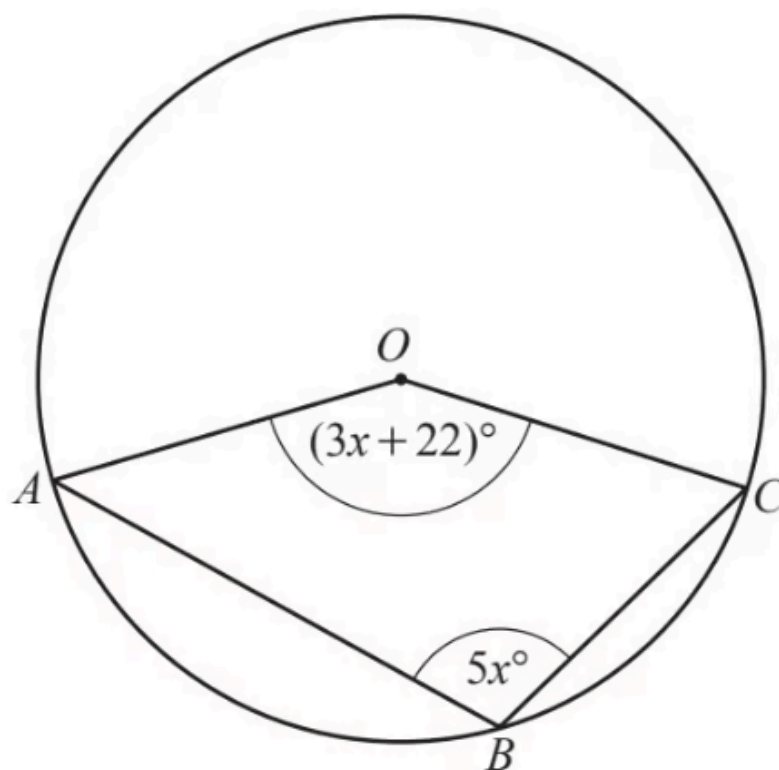
NOT TO
SCALE

The diagonals of the cyclic quadrilateral $ABCD$ intersect at X .

Explain why triangle ADX is similar to triangle BCX .

Give a reason for each statement you make.

(3 marks)



NOT TO
SCALE

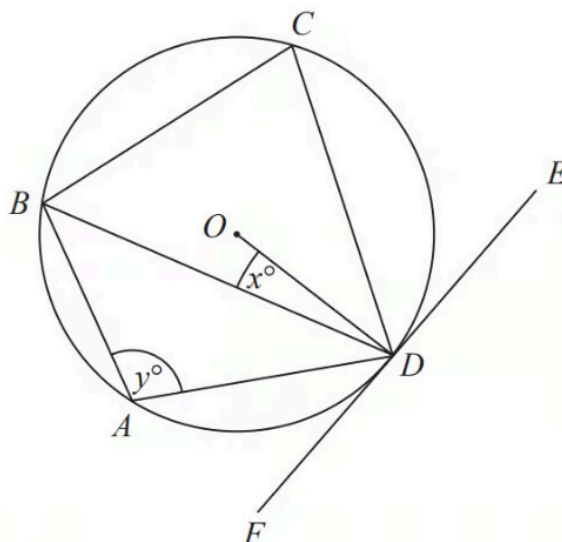
A , B and C lie on the circle, centre O .
Angle $AOC = (3x + 22)^\circ$ and angle $ABC = 5x^\circ$.

Find the value of x .

$x = \dots\dots\dots$

(4 marks)

4 (a)



A , B , C and D are points on the circumference of a circle, centre O .
 FDE is a tangent to the circle.

Show that $y - x = 90$

You must give a reason for each stage of your working.

(3 marks)

(b) Dylan was asked to give some possible values for x and y .

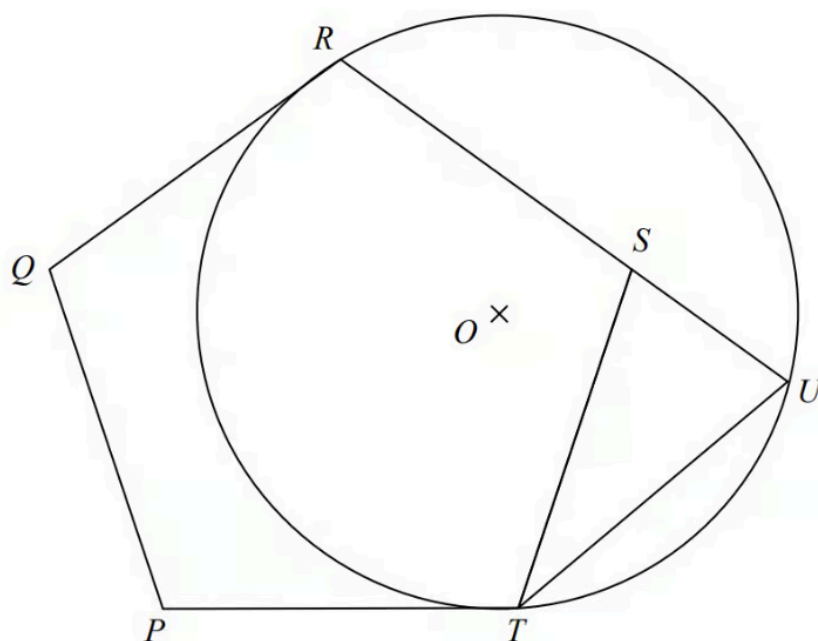
He said, " y could be 200 and x could be 110, because $200 - 110 = 90$ "

Is Dylan correct?

You must give a reason for your answer.

(1 mark)

5



$PQRST$ is a regular pentagon.
 R , U and T are points on a circle, centre O .
 QR and PT are tangents to the circle.
 RSU is a straight line.

Prove that $ST = UT$.

(5 marks)